

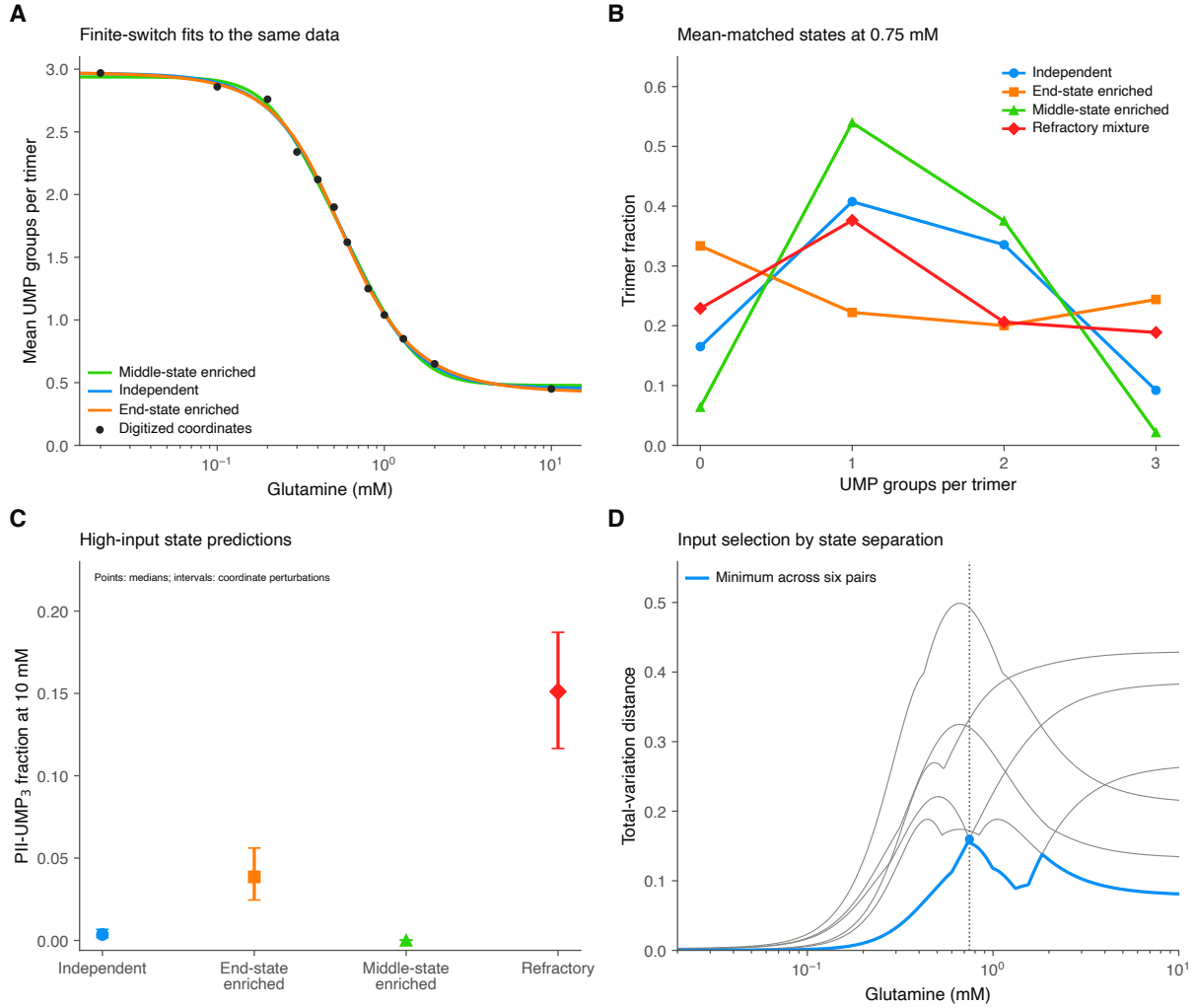


Figure S4: Data-matched mechanisms and discriminating PII states. (A) Separate four-parameter finite-switch fits at three fixed ladder shapes. (B) State distributions from exact mean-matched alternatives at 0.747 mM glutamine, including a fully modified refractory fraction. These inverse-constructed models differ from the finite-switch fits in A. Connecting lines guide the eye between discrete states. (C) Predicted fully uridylylated fractions at 10 mM. Points and intervals are medians and central 95% ranges from 300 shared coordinate perturbations; they exclude uncertainty in the assumed model structures and future measurement errors. (D) Pairwise total-variation distances between predicted distributions (gray); the dark curve is the smallest of the six distances. Its maximum on the specified 401-point grid is 0.160 at 0.747 mM. This is a candidate-dependent separation criterion, not an experimentally validated optimal design.